4.2.20

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Question

Find the direction and normal vector for the line $x + 2y = 6$

Theoretical Solution

Define the vectors:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad c = 6 \tag{1}$$

The line equation can be expressed as the dot product:

$$\mathbf{n}^\top \mathbf{x} = c \tag{2}$$

where $\mathbf{n}$ is the normal vector.

Direction Vector

The direction vector $\boldsymbol{m}$ is perpendicular to the normal vector $\boldsymbol{n}$. It can be taken as:

$$\boldsymbol{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \tag{3}$$

This follows since, if the direction vector is given by

$$\boldsymbol{m} = \begin{pmatrix} 1 \\ m \end{pmatrix} \tag{4}$$

then the normal vector is

$$\boldsymbol{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}. \tag{5}$$

Verification of Orthogonality

Verify that $\boldsymbol{n}$ and $\boldsymbol{m}$ are orthogonal:

$$\boldsymbol{n}^\top \boldsymbol{m} = 0 \tag{6}$$

Calculate:

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 0 \tag{7}$$

which confirms orthogonality.

Final Answer

1. Normal vector: $\boldsymbol{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$
2. Direction vector: $\boldsymbol{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

C Code

```
// Normal vector components
double normal[2] = {1, 2};

// Direction vector components (perpendicular to
    normal)
double direction[2] = {2, -1};

// Function to get normal vector pointer
double* get_normal() {
    return normal;
}

// Function to get direction vector pointer
double* get_direction() {
    return direction;
}
```

Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Line equation: x + 2y = 6
# Solve for y: y = (6 - x)/2

x = np.linspace(0, 7, 100)
y = (6 - x)/2

# Normal vector
n = np.array([1, 2])
# Direction vector (perpendicular to normal)
d = np.array([2, -1])

# Plot line
plt.plot(x, y, label='Line: x + 2y = 6')
```

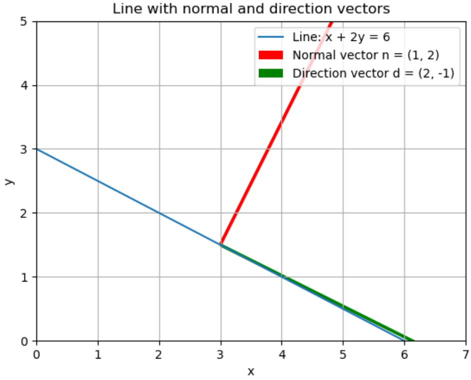
Python Code (continued)

```
# Plot normal vector starting from point (3, 1.5) on
    the line (3 + 2*1.5 = 6)
origin = np.array([3, 1.5])

plt.quiver(*origin, *n, color='red', scale=3, label='
    Normal vector n = (1, 2)')
plt.quiver(*origin, *d, color='green', scale=3, label=
    'Direction vector d = (2, -1)')

plt.xlim(0, 7)
plt.ylim(0, 5)
plt.xlabel('x')
plt.ylabel('y')
plt.title('Line with normal and direction vectors')
plt.grid(True)
plt.legend()
plt.show()
```

Plot


Line with normal and direction vectors
Line: x + 2y = 6
Normal vector n = (1, 2)
Direction vector d = (2, -1)
x
y
0
1
2
3
4
5
6
7